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Problem set: *L02 — Class 2: Introduction to statistical learning (textbook exercises)*

Instructions. These are the textbook exercises listed under *Practice* for Class 2 in `Map/class-plan.md`, grouped by book in the order of the class plan: ISLP (Block 1), Wooldridge (Block 2), Morettin & Singer (Block 3). Each statement is reproduced verbatim from the book PDF, with a source line; the Portuguese statements of Morettin & Singer stay in Portuguese. Computer exercises are solved in Python (numpy, pandas, statsmodels; no scikit-learn): the answer key shows the code that matters and the key output, and `Listas/L02-book-exercises-check.py` reproduces and asserts every number printed in it. Data: `Listas/data/` (provenance in `Listas/data/README.md`) and the `wooldridge` Python package. Show every step.

Block 1 — *ISLP*, James et al. (2023) *(Ch. 2, 5, 7, 4 • 12 problems)*

Question 1. Flexible versus inflexible methods (ISLP 2.1).

Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 2.1 (Section 2.4, Conceptual), p. 63. Reproduced verbatim.

For each of parts (a) through (d), indicate whether we would generally expect the performance of a flexible statistical learning method to be better or worse than an inflexible method. Justify your answer.

(a) The sample size n is extremely large, and the number of predictors p is small.

(b) The number of predictors p is extremely large, and the number of observations n is small.

(c) The relationship between the predictors and response is highly non-linear.

(d) The variance of the error terms, i.e. $\sigma^2 = \text{Var}(\epsilon)$, is extremely high.

Solution:

Question 2. Sketch of the bias–variance decomposition (ISLP 2.3).

Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 2.3 (Section 2.4, Conceptual), p. 63. Reproduced verbatim.

We now revisit the bias-variance decomposition.

(a) Provide a sketch of typical (squared) bias, variance, training error, test error, and Bayes (or irreducible) error curves, on a single plot, as we go from less flexible statistical learning methods towards more flexible approaches. The x-axis should represent the amount of flexibility in the method, and the y-axis should represent the values for each curve. There should be five curves. Make sure to label each one.

(b) Explain why each of the five curves has the shape displayed in part (a).

Solution:

Question 3. k-fold cross-validation (ISLP 5.3).
Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 5.3 (Section 5.4, Conceptual), p. 225. Reproduced verbatim.

We now review k-fold cross-validation.

(a) Explain how k-fold cross-validation is implemented.

(b) What are the advantages and disadvantages of k-fold cross-validation relative to:

i. The validation set approach?

ii. LOOCV?

Solution:

Question 4. Validation-set error of a logistic regression on `Default` (ISLP 5.5).
Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 5.5 (Section 5.4, Applied), pp. 225–226. Reproduced verbatim.

In Chapter 4, we used logistic regression to predict the probability of `default` using `income` and `balance` on the `Default` data set. We will now estimate the test error of this logistic regression model using the validation set approach. Do not forget to set a random seed before beginning your analysis.

(a) Fit a logistic regression model that uses `income` and `balance` to predict `default`.

(b) Using the validation set approach, estimate the test error of this model. In order to do this, you must perform the following steps:

i. Split the sample set into a training set and a validation set.

ii. Fit a multiple logistic regression model using only the training observations.

iii. Obtain a prediction of default status for each individual in the validation set by computing the posterior probability of default for that individual, and classifying the individual to the `default` category if the posterior probability is greater than 0.5.

iv. Compute the validation set error, which is the fraction of the observations in the validation set that are misclassified.

(c) Repeat the process in (b) three times, using three different splits of the observations into a training set and a validation set. Comment on the results obtained.

(d) Now consider a logistic regression model that predicts the probability of `default` using `income`, `balance`, and a dummy variable for `student`. Estimate the test error for this model using the validation set approach. Comment on whether or not including a dummy variable for `student` leads to a reduction in the test error rate.

Solution:

Question 5. LOOCV for the polynomial degree on simulated data (ISLP 5.8).
Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 5.8 (Section 5.4, Applied), p. 227. Reproduced verbatim.

We will now perform cross-validation on a simulated data set.

(a) Generate a simulated data set as follows:

```
rng = np.random.default_rng(1)
x = rng.normal(size=100)
y = x - 2 * x**2 + rng.normal(size=100)
```

In this data set, what is n and what is p? Write out the model used to generate the data in equation form.

(b) Create a scatterplot of X against Y. Comment on what you find.

(c) Set a random seed, and then compute the LOOCV errors that result from fitting the following four models using least squares:

i. $Y = \beta_0 + \beta_1 X + \epsilon$
ii. $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \epsilon$
iii. $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon$
iv. $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 X^4 + \epsilon.$

Note you may find it helpful to use the `data.frame()` function to create a single data set containing both X and Y.

(d) Repeat (c) using another random seed, and report your results. Are your results the same as what you got in (c)? Why?

(e) Which of the models in (c) had the smallest LOOCV error? Is this what you expected? Explain your answer.

(f) Comment on the statistical significance of the coefficient estimates that results from fitting each of the models in (c) using least squares. Do these results agree with the conclusions drawn based on the cross-validation results?

Solution:

Question 6. CV-chosen polynomial degree and step functions for `Wage` (ISLP 7.6).
Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 7.6 (Section 7.9, Applied), p. 327. Reproduced verbatim.

In this exercise, you will further analyze the `Wage` data set considered throughout this chapter.

(a) Perform polynomial regression to predict `wage` using `age`. Use cross-validation to select the optimal degree d for the polynomial. What degree was chosen, and how does this compare to the results of hypothesis testing using ANOVA? Make a plot of the resulting polynomial fit to the data.

(b) Fit a step function to predict `wage` using `age`, and perform cross-validation to choose the optimal number of cuts. Make a plot of the fit obtained.

Solution:

Question 7. Polynomials and regression splines for `nox` on `dis` (ISLP 7.9).
Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 7.9 (Section 7.9, Applied), p. 327. Reproduced verbatim.

This question uses the variables `dis` (the weighted mean of distances to five Boston employment centers) and `nox` (nitrogen oxides concentration in parts per 10 million) from the `Boston` data. We will treat `dis` as the predictor and `nox` as the response.

(a) Use the `poly()` function from the `ISLP.models` module to fit a cubic polynomial regression to predict `nox` using `dis`. Report the regression output, and plot the resulting data and polynomial fits.

(b) Plot the polynomial fits for a range of different polynomial degrees (say, from 1 to 10), and report the associated residual sum of squares.

(c) Perform cross-validation or another approach to select the optimal degree for the polynomial, and explain your results.

(d) Use the `bs()` function from the `ISLP.models` module to fit a regression spline to predict `nox` using `dis`. Report the output for the fit using four degrees of freedom. How did you choose the knots? Plot the resulting fit.

(e) Now fit a regression spline for a range of degrees of freedom, and plot the resulting fits and report the resulting RSS. Describe the results obtained.

(f) Perform cross-validation or another approach in order to select the best degrees of freedom for a regression spline on this data. Describe your results.

Solution:

Question 8. Logistic function and logit are equivalent (ISLP 4.1).
Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 4.1 (Section 4.8, Conceptual), p. 193. Reproduced verbatim. Equations (4.2) and (4.3) of the book are restated in the solution.

Using a little bit of algebra, prove that (4.2) is equivalent to (4.3). In other words, the logistic function representation and logit representation for the logistic regression model are equivalent.

Solution:

Question 9. Odds and probabilities (ISLP 4.9).
Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 4.9 (Section 4.8, Conceptual), p. 195. Reproduced verbatim.

This problem has to do with *odds*.

(a) On average, what fraction of people with an odds of 0.37 of defaulting on their credit card payment will in fact default?

(b) Suppose that an individual has a 16 % chance of defaulting on her credit card payment. What are the odds that she will default?

Solution:

Question 10. Logistic regression versus the softmax coding (ISLP 4.12).
Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 4.12 (Section 4.8, Conceptual), p. 196. Reproduced verbatim, including a misprint in item (d) flagged in the note after the statement.

Suppose that you wish to classify an observation $X \in \mathbb{R}$ into `apples` and `oranges`. You fit a logistic regression model and find that

$$\widehat{\Pr}(Y = \texttt{orange} \mid X = x) = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 x)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)}.$$

Your friend fits a logistic regression model to the same data using the *softmax* formulation in (4.13), and finds that

$$\widehat{\Pr}(Y = \texttt{orange} \mid X = x) = \frac{\exp(\hat{\alpha}_{\texttt{orange0}} + \hat{\alpha}_{\texttt{orange1}} x)}{\exp(\hat{\alpha}_{\texttt{orange0}} + \hat{\alpha}_{\texttt{orange1}} x) + \exp(\hat{\alpha}_{\texttt{apple0}} + \hat{\alpha}_{\texttt{apple1}} x)}.$$

(a) What is the log odds of `orange` versus `apple` in your model?

(b) What is the log odds of `orange` versus `apple` in your friend's model?

(c) Suppose that in your model, $\hat{\beta}_0 = 2$ and $\hat{\beta}_1 = -1$. What are the coefficient estimates in your friend's model? Be as specific as possible.

(d) Now suppose that you and your friend fit the same two models on a different data set. This time, your friend gets the coefficient estimates $\hat{\alpha}_{\texttt{orange0}} = 1.2$, $\hat{\alpha}_{\texttt{orange1}} = -2$, $\hat{\alpha}_{\texttt{orange0}} = 3$, $\hat{\alpha}_{\texttt{orange1}} = 0.6$. What are the coefficient estimates in your model?

(e) Finally, suppose you apply both models from (d) to a data set with 2,000 test observations. What fraction of the time do you expect the predicted class labels from your model to agree with those from your friend's model? Explain your answer.

Note (not in the book): in (d) the book prints the second pair as $\hat{\alpha}_{\texttt{orange0}} = 3$, $\hat{\alpha}_{\texttt{orange1}} = 0.6$; it is a misprint for $\hat{\alpha}_{\texttt{apple0}} = 3$, $\hat{\alpha}_{\texttt{apple1}} = 0.6$, and is read that way below.

Solution:

Question 11. Classifying the direction of the market with `Weekly` (ISLP 4.13).

Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 4.13 (Section 4.8, Applied), pp. 196–197. Reproduced verbatim.

This question should be answered using the `Weekly` data set, which is part of the `ISLP` package. This data is similar in nature to the `Smarket` data from this chapter's lab, except that it contains 1,089 weekly returns for 21 years, from the beginning of 1990 to the end of 2010.

(a) Produce some numerical and graphical summaries of the `Weekly` data. Do there appear to be any patterns?

(b) Use the full data set to perform a logistic regression with `Direction` as the response and the five lag variables plus `Volume` as predictors. Use the summary function to print the results. Do any of the predictors appear to be statistically significant? If so, which ones?

(c) Compute the confusion matrix and overall fraction of correct predictions. Explain what the confusion matrix is telling you about the types of mistakes made by logistic regression.

(d) Now fit the logistic regression model using a training data period from 1990 to 2008, with `Lag2` as the only predictor. Compute the confusion matrix and the overall fraction of correct predictions for the held out data (that is, the data from 2009 and 2010).

(e) Repeat (d) using LDA.

(f) Repeat (d) using QDA.

(g) Repeat (d) using KNN with $K = 1$.

(h) Repeat (d) using naive Bayes.

(i) Which of these methods appears to provide the best results on this data?

(j) Experiment with different combinations of predictors, including possible transformations and interactions, for each of the methods. Report the variables, method, and associated confusion matrix that appears to provide the best results on the held out data. Note that you should also experiment with values for K in the KNN classifier.

Solution:

Question 12. High versus low gas mileage with `Auto` (ISLP 4.14).
Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 4.14 (Section 4.8, Applied), pp. 197–198. Reproduced verbatim.

In this problem, you will develop a model to predict whether a given car gets high or low gas mileage based on the `Auto` data set.

(a) Create a binary variable, `mpg01`, that contains a 1 if `mpg` contains a value above its median, and a 0 if `mpg` contains a value below its median. You can compute the median using the `median()` method of the data frame. Note you may find it helpful to add a column `mpg01` to the data frame by assignment. Assuming you have stored the data frame as `Auto`, this can be done as follows:

```
Auto['mpg01'] = mpg01
```

(b) Explore the data graphically in order to investigate the association between `mpg01` and the other features. Which of the other features seem most likely to be useful in predicting `mpg01`? Scatterplots and boxplots may be useful tools to answer this question. Describe your findings.

(c) Split the data into a training set and a test set.

(d) Perform LDA on the training data in order to predict `mpg01` using the variables that seemed most associated with `mpg01` in (b). What is the test error of the model obtained?

(e) Perform QDA on the training data in order to predict `mpg01` using the variables that seemed most associated with `mpg01` in (b). What is the test error of the model obtained?

(f) Perform logistic regression on the training data in order to predict `mpg01` using the variables that seemed most associated with `mpg01` in (b). What is the test error of the model obtained?

(g) Perform naive Bayes on the training data in order to predict `mpg01` using the variables that seemed most associated with `mpg01` in (b). What is the test error of the model obtained?

(h) Perform KNN on the training data, with several values of K, in order to predict `mpg01`. Use only the variables that seemed most associated with `mpg01` in (b). What test errors do you obtain? Which value of K seems to perform the best on this data set?

Solution:

Block 2 — Wooldridge (2016), 6th ed. *(Ch. 17 • 3 problems)*

Question 13. Point spreads: linear probability model versus probit (W 17.C1).
Source: Wooldridge (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 17, Computer Exercise C1, p. 560. Reproduced verbatim.

Use the data in PNTSPRD for this exercise.

(i) The variable *favwin* is a binary variable if the team favored by the Las Vegas point spread wins. A linear probability model to estimate the probability that the favored team wins is

$$P(favwin = 1 \mid spread) = \beta_0 + \beta_1 spread.$$

Explain why, if the spread incorporates all relevant information, we expect $\beta_0 = .5$.

(ii) Estimate the model from part (i) by OLS. Test $H_0\colon \beta_0 = .5$ against a two-sided alternative. Use both the usual and heteroskedasticity-robust standard errors.

(iii) Is *spread* statistically significant? What is the estimated probability that the favored team wins when $spread = 10$?

(iv) Now, estimate a probit model for $P(favwin = 1 \mid spread)$. Interpret and test the null hypothesis that the intercept is zero. [*Hint:* Remember that $\Phi(0) = .5$.]

(v) Use the probit model to estimate the probability that the favored team wins when $spread = 10$. Compare this with the LPM estimate from part (iii).

(vi) Add the variables *favhome*, *fav25*, and *und25* to the probit model and test joint significance of these variables using the likelihood ratio test. (How many df are in the chi-square distribution?) Interpret this result, focusing on the question of whether the spread incorporates all observable information prior to a game.

Solution:

Question 14. Quadratics in the Poisson regression for arrests (W 17.C4).

Source: Wooldridge (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 17, Computer Exercise C4, p. 561. Reproduced verbatim.

In Example 9.1, we added the quadratic terms $pcnv^2$, $ptime86^2$, and $inc86^2$ to a linear model for *narr86*.

(i) Use the data in CRIME1 to add these same terms to the Poisson regression in Example 17.3.

(ii) Compute the estimate of σ^2 given by $\hat{\sigma}^2 = (n - k - 1)^{-1} \sum_{i=1}^{n} \hat{u}_i^2 / \hat{y}_i$. Is there evidence of overdispersion? How should the Poisson MLE standard errors be adjusted?

(iii) Use the results from parts (i) and (ii) and Table 17.5 to compute the quasi-likelihood ratio statistic for joint significance of the three quadratic terms. What do you conclude?

Solution:

Question 15. Poisson regression for fertility (W 17.C5).

Source: Wooldridge (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 17, Computer Exercise C5, p. 561. Reproduced verbatim.

Refer to Table 13.1 in Chapter 13. There, we used the data in FERTIL1 to estimate a linear model for *kids*, the number of children ever born to a woman.

(i) Estimate a Poisson regression model for *kids*, using the same variables in Table 13.1. Interpret the coefficient on *y82*.

(ii) What is the estimated percentage difference in fertility between a black woman and a nonblack woman, holding other factors fixed?

(iii) Obtain $\hat{\sigma}$. Is there evidence of over- or underdispersion?

(iv) Compute the fitted values from the Poisson regression and obtain the R-squared as the squared correlation between $kids_i$ and $\widehat{kids}_i$. Compare this with the R-squared for the linear regression model.

Solution:

Block 3 — Morettin & Singer (2021) *(Ch. 8 • 4 problems (in Portuguese))*

Question 16. KNN with LOOCV for the disc data (MS 8.9).

Source: Morettin & Singer (2021), *Estatística e Ciência de Dados* (versão preliminar, abril 2021), Capítulo 8, Exercício 9 (Seção 8.7, Exercícios), p. 289. Reproduced verbatim (Portuguese original).

Use o conjunto de dados do Exemplo 8.1 e distância aberta como variável preditora. Use LOOCV e o classificador KNN, com vizinhos mais próximos de 1 a 5.

a) Qual o melhor classificador baseado na acurácia?

b) Obtenha a matriz de confusão e realize o teste de McNemar.

c) Obtenha a sensibilidade e a especificidade e explique seus significados nesse caso.

Solution:

Question 17. LDA with two inhibin predictors and k-fold CV (MS 8.4).

Source: Morettin & Singer (2021), *Estatística e Ciência de Dados* (versão preliminar, abril 2021), Capítulo 8, Exercício 4 (Seção 8.7, Exercícios), p. 289. Reproduced verbatim (Portuguese original; "varáveis" is the book's spelling).

Reanalise os dados do Exemplo 8.3 considerando duas varáveis preditoras `inibpre` e `inibpos` e validação cruzada de ordem k com diferentes valores de k. Avalie o efeito da semente nos resultados.

Solution:

Question 18. Choosing a logistic model for endometriosis by 5-fold CV (MS 8.6).
Source: Morettin & Singer (2021), *Estatística e Ciência de Dados* (versão preliminar, abril 2021), Capítulo 8, Exercício 6 (Seção 8.7, Exercícios), p. 289. Reproduced verbatim (Portuguese original).

Considere os dados do arquivo `endometriose2`. Com propósitos de previsão, ajuste modelos de regressão logística com validação cruzada de ordem 5 tendo `endometriose` como variável resposta e

i) `idade`, `dormenstrual`, `dismenorreia` ou `tipoesteril` separadamente como variáveis explicativas (4 modelos);

ii) cada par das variáveis do item i) como variáveis explicativas (6 modelos);

iii) cada trio das variáveis do item i) como variáveis explicativas (4 modelos);

iv) as quatro variáveis do item i) como variáveis explicativas (1 modelo).

Escolha o melhor modelo com base na acurácia.

Solution:

Question 19. Logistic regression, Bayes classifier and Fisher's discriminant (MS 8.11).
Source: Morettin & Singer (2021), *Estatística e Ciência de Dados* (versão preliminar, abril 2021), Capítulo 8, Exercício 11 (Seção 8.7, Exercícios), p. 289. Reproduced verbatim (Portuguese original). Exercise 5, to which it refers, is quoted for context.

Para os dados do Exercício 5, compare os classificadores usando regressão logística, classificador bayesiano e função discriminante linear de Fisher.

Exercise 5 (p. 289): Analise os dados do arquivo `endometriose2` por meio das técnicas de classificação descritas neste capítulo, utilizando validação cruzada com diferentes parâmetros. Compare-as e discuta os resultados.

Solution: